

The empirical recurrence for OEIS A283639: recovering a conjecture that is not written down

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Abstract

OEIS A283639 records “Empirical recurrence of order 96”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 410$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 96, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 96 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A283639 is “Number of $6 \times n$ 0..1 arrays with no 1 equal to more than one of its horizontal, diagonal and antidiagonal neighbors, with the exception of exactly two elements.”. It has offset 1 and begins

0, 606, 28828, 856710, 20051838, 490925804, 10666178322, 218881780434, ...

The entry states:

Empirical recurrence of order 96 (see link above)

As of the “Last modified” line on the live entry (Mar 12 2017 14:40:10, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 5113$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 410$ of the 5113, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 410$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 820$ exact terms of the model returns order 96 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{96} c_i a(n-i),$$

c_1	12	c_{25}	106112128586891098989	c_{49}	4096056910543805295084106032
c_2	288	c_{26}	−449981244502794801000	c_{50}	−2718227876430196739607313593
c_3	376	c_{27}	1703231808233930919687	c_{51}	−6707517763260247424684695726
c_4	−52692	c_{28}	−6015880835117088588990	c_{52}	−24716534155111221386881106049
c_5	−476694	c_{29}	20138394258871009717599	c_{53}	−26118524749148568020469994212
c_6	73909	c_{30}	−59697461516908919182097	c_{54}	−31527054864458105439192715277
c_7	41432517	c_{31}	168573164082202091593857	c_{55}	14355242812498233794724743655
c_8	385020264	c_{32}	−428938196830617929749686	c_{56}	57779957924100480866507216442
c_9	2325159364	c_{33}	991066009312593497387036	c_{57}	159176892369929335521976725138
c_{10}	8732289954	c_{34}	−2181166685671891689356022	c_{58}	193427750072117050016148574473
c_{11}	30988036470	c_{35}	4117199828404893976745838	c_{59}	189248346016581471463160259675
c_{12}	−18855792660	c_{36}	−7565047985203620700959997	c_{60}	−8222786683978661482912358661
c_{13}	−262118106477	c_{37}	12584041652241177988635735	c_{61}	−222859118844781823846834978370
c_{14}	−4139521192839	c_{38}	−17377063614380325927658500	c_{62}	−547675852047502417934012415498
c_{15}	−3381739764036	c_{39}	31089593939564219303231657	c_{63}	−740552875935611062891339961986
c_{16}	−81408059140290	c_{40}	−35397289807880399381149872	c_{64}	−798924684156990139774915660338
c_{17}	255718360099569	c_{41}	61349264964147429477833664	c_{65}	−303739368544344551395943174796
c_{18}	−1314275443878439	c_{42}	−136511259613460196417939930	c_{66}	376412018237414321169921601986
c_{19}	10914602552598090	c_{43}	77929607820033633572436444	c_{67}	1125920526468647118551809175040
c_{20}	−41504872401216540	c_{44}	−440867241426219380193180360	c_{68}	1578135477379794010379776190640
c_{21}	265953574459888775	c_{45}	317763534971923093455515676	c_{69}	1857259889853464503322280918444
c_{22}	−1193500889246321952	c_{46}	−186768686100221138273514063	c_{70}	1445659890226038406359846150024
c_{23}	5473478741632205790	c_{47}	2150135341295179797942879468	c_{71}	232428989718526873779522677916
c_{24}	−26002934963607105672	c_{48}	1653536006394909180484121778	c_{72}	−1120978626101302954966547699076

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 96, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $96 + 4$ terms removed returns order 96 as well, so $\deg q_{\min} = 96 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $96 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 96 that this sequence satisfies — and that family turns out to have exactly one member.

References

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